THE IMPACT OF OIL AND GAS EXTRACTION ON INFANT HEALTH

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ABSTRACT

The benefits and costs of resource extraction are currently being hotly debated in the case of unconventional natural gas development (commonly known as "fracking"). Colorado provides a unique research environment to study the health impacts of conventional and unconventional forms of oil and gas development (OGD) given its long history of conventional OGD and recent expansion of unconventional OGD. I utilize detailed vital statistics and the mother's residential address to define proximity to drilling activity. Using a difference-in-differences model that compares mothers residing within 1 km to mothers residing 1–5 km of a well (before and during/after drilling), I find that proximity to wells before birth increases adverse birth outcomes and pregnancy-related complications. I find impacts on the extensive and intensive margins and across well types. I also present results suggesting no impacts on fertility or mobility. These results are robust to multiple specifications and suggest that policies to mitigate against the risks of living near oil and gas development may be warranted.

KEYWORDS: infant health, oil and gas development, air pollution, low birth weight
JEL CLASSIFICATION: I10, Q33, Q53

I. Introduction

One of the central innovations of the oil and gas industry in the United States is the use of high-volume horizontal hydraulic fracturing (HVHF) to gain access to formerly inaccessible formations, such as unconventional sources like shale. The "Shale Revolution," as it has been coined, has increased domestic production by 88 percent for natural gas and 137 percent for crude oil from 2005 to 2019 (DOE 2021). Owing to public attention and policy interest in assessing the health impacts of shale gas development, there is a growing body of literature in economics that has found adverse birth outcomes for communities living near unconventional drilling (Currie, Greenstone, and Meckel 2017; Hill 2018; Hill

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and Ma 2022; Willis et al. 2022). Much of the public fear, and recent academic research, centers around the recent growth in unconventional sources, such as shale gas, but mostly ignores more conventional forms of oil and gas development (OGD).[1] This paper fills this gap in the literature by studying the effects of both conventional and unconventional OGD on infant health in a state with multiple types of resources and development activities.

In this paper, I leverage the expansion of oil, gas, and coalbed methane development, including vertical and horizontal wells in Colorado, to provide new quasi-experimental evidence of the infant health impacts of resource extraction. In addition to the growing economic literature, there is also a large epidemiological literature assessing associations between communities with drilling and birth outcomes (see Deziel et al. [2020] for a recent review). One of these studies found protective effects for preterm birth and low birth weight in Colorado and increased risk of birth defects (McKenzie et al. 2014). I find that there is positive selection into exposure to drilling in Colorado that may explain these counterintuitive findings. Mothers living within 1 km of drilling have higher income and are more highly educated, more likely to be married, less likely to smoke, less likely to be teenaged, and less likely to be on Medicaid. This selection may be explained by siting of drilling in particular counties in Colorado that are also population centers or where industry can lease large amounts of property. Given these level differences (and no evidence of pre-trends), I use a difference-in-differences research design to compare birth outcomes for maternal residences close (i.e., 0–1 km from a well) to slightly farther away (1–5 km from a well), before and during/after drilling to estimate the average effect of living in close proximity to any well during pregnancy. The parallel trends assumption is tested by examining the observable characteristics of the mothers and pre-trends in the outcomes of interest across time and distance. A range of specifications are estimated in an effort to provide evidence that the research design is robust. I also estimate models of well density, mother fixed effects, well fixed effects, extreme health outcomes, and maternal health outcomes and show robustness to the inclusion of controls for maternal income and insurance status.

I find both statistically and economically significant effects on infant and maternal health when pooling all well types (oil, gas, coalbed methane; vertical and horizontal). I find that for the extensive margin, birth weight decreased and low birth weight (LBW) prevalence increased, on average. I also find impacts on gestational age and the prevalence of preterm birth (PTB). For LBW and PTB the magnitude of the impacts is ~30 percent relative to the pre-drilling mean. This effect size is similar in magnitude to other studies on shale gas development (Currie, Greenstone, and Meckel 2017; Hill 2018). I find consistent evidence as well for intensive-margin models. For example, an additional well drilled within 1 km reduces birth weight by 2.5 grams, on average. I test sensitivity to spatial fixed effects (zip code, county), maternal fixed effects, and well fixed effects, and the results are qualitatively robust. I also find an increase in gestational diabetes and hypertension for mothers, which is consistent with one other study in Texas (Willis et al. 2022). The results

1 There is economic literature studying the 1970s oil boom, but this literature has focused on labor market outcomes and not infant health (Jacobsen and Parker 2016; Jacobsen, Parker, and Winikoff 2021).

are also robust to multiple specification checks, changes in temporal and spatial sample restrictions (e.g., no spatial restriction, within 15 km), and placebo regressions.

One major threat to my identification is that OGD leads to community and local economic changes that could in turn impact birth outcomes either through compositional changes or through economic shocks. A large and growing literature in economics has shown that OGD has impacts on housing values (Muehlenbachs, Spiller, and Timmins 2014; Gopalakrishnan and Klaiber 2014; Bennett and Loomis 2015), employment (Weber 2012; Marchand 2012; DeLeire, Eliason, and Timmins 2014; Wrenn, Kelsey, and Jaenicke 2015; Feyrer, Mansur, and Sacerdote 2020), royalty payments and wage income (Hardy and Kelsey 2015; Brown, Fitzgerald, and Weber 2016; Feyrer, Mansur, and Sacerdote 2020), educational response (Cascio and Narayan 2022), and local governmental finance (Newell and Raimi 2015). Other studies have shown impacts on migration (Wilson 2022), fertility (Kearney and Wilson 2018), and teen pregnancy (Owen 2022).[2] Consistent with this literature, I find a 50 percent increase in the likelihood of teen motherhood in these communities. When I remove these mothers, my results hold, indicating that my main results are not driven by this increase. I also find some suggestive evidence that mothers are more likely to move after drilling for lower-order births (between first to second and between second to third births) and less likely to move for higher-order births (third to fourth births). Overall, I find modest impacts on mobility, with around a 1–4 percent increased likelihood of mothers ever moving in response to drilling (not statistically precise). When I limit the analysis to mothers who never move, my main results are robust. Finally, I investigate fertility impacts and do not find consistent evidence of a large fertility impact, which is inconsistent with the prior literature (Kearney and Wilson 2018).[3]

Finally, given that my main findings pool all wells drilled prior to birth, I use interaction models to understand whether certain types of drilling may be driving the overall effects. First, I interact with wells that are producing prior to pregnancy to understand whether results are driven by newly drilled wells or by older producing wells. I find that most of the effect is driven by new wells, but producing wells have a small additional impact that is not statistically significant. Second, I estimate the effect of horizontal wells versus vertical wells and find that there is no statistical difference between these well types in Colorado. One other study has found slightly larger impacts for horizontal drilling in Texas (Willis et al. 2022), so this finding may show that vertical drilling in Colorado is not the same as what occurs in other states. Finally, I compare coalbed methane, oil, and natural gas wells and find the largest impacts for coalbed methane, but adverse impacts for all three well types. These heterogeneity analyses indicate that all types of extraction activities are associated with adverse birth outcomes and that perhaps the policy debate should also include conventional natural gas, oil, and other resource types such as coalbed methane.

2 See Black et al. (2021) for a recent review of the economics literature studying the economic, environmental, and health impacts of fracking.

3 I aggregate to county, and Kearney and Wilson (2018) aggregate to the commuting zone, which may explain the inconsistency. My study is at an even finer spatial scale of within 5 km. I rule out fertility explaining the main results.

The economic benefit of drilling for natural gas is potentially huge for landowners, states, and industry. Many have written about the promise of this energy boom to improve the US economy (Aguilera and Radetzki 2013; Ames et al. 2012; Linn, Muehlenbachs, and Wang 2014; Mason, Muehlenbachs, and Olmstead 2015; Hausman and Kellogg 2015; Bartik et al. 2019; Feyrer, Mansur, and Sacerdote 2020) and potentially public health from the reduction in the use of coal for electricity production (Johnsen, LaRiviere, and Wolff 2019). Any benefit-cost analysis requires taking into account the local external costs of drilling activities, and this paper provides robust estimates related to the health costs for birth outcomes in affected communities.

II. Oil and Gas Development in Colorado

A. DRILLING AND PRODUCTION OVERVIEW

Colorado has had a long history of resource extraction, with the first wells drilled in the 19th century and more recent development being regulated and managed by the Colorado Oil and Gas Conservation Commission (COGCC) starting in the 1950s. As of 2012, Colorado was the sixth largest producer of natural gas and nineth largest producer of oil. From 2007 to 2011 (the last year of this study), oil and natural gas production increased 64 percent and 27 percent, respectively. Online Appendix Figure A1 shows trends over this period by resource type (e.g., wet gas, dry gas, coalbed methane or coal gas, and oil) and shows trends in the utilization of horizontal drilling. Much of the increase in horizontal drilling began after 2005 across all resource extraction activities. There is a large amount of conventional, or vertical, drilling that occurs during this period, motivating the inclusion of these types of wells in this study.

B. OIL AND GAS DEVELOPMENT AS A POTENTIAL POLLUTION SOURCE

Pathways for human exposure (e.g., air, water) from OGD and high-volume hydraulic fracturing (HVHF) have become more well-defined in the peer-reviewed literature as of late. HVHF operations have been linked to surface and groundwater contamination (Hill and Ma 2017, 2022; Bonetti, Leuz, and Michelon 2021), and emissions of harmful air pollutants occur throughout the life cycle of HVHF development (Zhang et al. 2023). For both water and air pollution, the economic literature has found that most of the impacts occur during the drilling and hydraulic fracturing phases, with smaller amounts of pollution occurring during the production phase. Other studies have found increased light pollution following development that could also have adverse effects on pregnancy outcomes (Boslett et al. 2021; Smith and Wills 2018). See these papers and a recent review by Black et al. (2021) that describe in detail the environmental mechanisms of exposure.

C. REPRODUCTIVE HEALTH OUTCOMES AND UNCONVENTIONAL DRILLING

Using the above literature review of potential contaminants, I searched the economic and epidemiological literature to better understand whether these contaminants are likely to impact reproductive health (as measured by low birth weight and premature birth). In Online Appendix Table A1, I provide a tabular literature review of pertinent studies associated with

each air pollutant. Most, if not all, contaminants are likely to contribute to the risk of adverse birth outcomes. The magnitude of the findings in this paper are well within the range of findings in these other studies.

Concurrently to the research reported here, there have been multiple studies in the epidemiology literature addressing birth outcomes to mothers who live in proximity to natural gas development (see Deziel et al. [2020] for a scoping review)[4] and three other studies in the economic literature based on unconventional (shale) gas development in Pennsylvania (Currie, Greenstone, and Meckel 2017; Hill 2018; Hill and Ma 2022). Most of these studies consistently find adverse birth outcomes for mothers living in communities with drilling.

III. Data Sources

My analysis is based upon a data set acquired from the Colorado Oil and Gas Conservation Commission (COGCC) that contains information on each well drilled in Colorado since 1985.[5] The data set contains information on location of well (latitude and longitude), spud date (timing that drilling began), first production date, test dates, total depth date (when vertical drilling was completed), directional drilling (also called horizontal), resource type (i.e., oil, gas, or coalbed methane), geological formation, company, and other details.

My second source of data comes from birth certificate records from the Colorado Department of Public Health and the Environment (CDPHE) from 2000 to 2011. The natality records contain detailed information on every birth in the state including health at birth and background information on the mother and father; background information includes race, education, and marital status, as well as prenatal care and whether the mother smoked or drank alcohol during her pregnancy. My study makes use of the mother's exact address (geocoded to latitude and longitude), which is merged to the oil and gas locations to define proximity to drilling.[6] Each birth is linked to all OGD wells within 30 km of the mother's residence with flags for whether the wells are drilled before conception, during pregnancy, or after birth. The nearest well drilled *prior to birth* is used to define exposure buffers for the treated group (i.e., exposed within 1 km prior to birth); however, because residences are exposed to multiple wells concurrently at different time points, the comparison group is defined by any well within each buffer. I further restrict my analysis to a sample where the nearest well is drilled within two years before or after the birth date.

I make use of four primary reproductive health outcomes.[7] Birth weight is defined as the weight of the infant measured at birth in grams. Gestation length is defined as the

4 Notably, a pair of studies in Texas employing difference-in-differences found increased risk of term birth weight (Willis et al. 2021) and hypertensive conditions of the mother (Willis et al. 2022).

5 These data are self-reported by industry; however, they are required to submit these data to the state. The data quality improves over time.

6 CDPHE performed this merge and released the data stripped of geographical indicators, except those I requested specifying distance to nearest wells within 30 km of each mother's residence.

7 Other outcomes that may be of interest, such as fetal/infant mortality and congenital anomalies, are very rare events. When restricting the data set to those very close to gas wells or permits, there are insufficient cases in Colorado for there to be a measurable effect for these outcomes.

number of weeks of gestation. Low birth weight (LBW), defined as birth weight less than 2,500 grams, is commonly used as a key indicator of infant health, and hence is one of the outcomes examined. Premature birth (also called preterm birth, PTB), defined as gestation length less than 37 weeks, is associated with a greater risk for short- and long-term complications, including disabilities and impediments to growth and mental development. I also explore different levels of low birth weight and premature birth (e.g., very LBW, extremely PTB).

A. DESCRIPTIVE STATISTICS

Table 1 provides summary statistics for the primary difference-in-differences analysis sample. In the difference-in-differences model, I compare residences within 1 km of a gas well with those 1–5 km from a gas well.[8] Most of the statistically significant differences between these two samples are actually not very economically important (e.g., they are small in magnitude). However, mother characteristics closest to wells (within 1 km) are indicative of better socioeconomic status versus those farther away (1–5 km). Those mothers who reside within 1 km are more educated, more white, less likely to smoke while pregnant, and more likely to be married. To look at these differences for births that occur before or after the nearest well is drilled, I also show distance gradients of maternal characteristics in Online Appendix Figure A2. These show a consistent relationship for births both before and after drilling near wells, which suggests that mothers with higher socioeconomic status are more likely to be exposed to drilling at close proximity. The difference in levels for maternal characteristics observed beyond 5 km are one reason I limit my analysis to within 5 km for my main results (and show robustness to this decision in the robustness checks in Section VI.D). I control for the observable characteristics of the mother in the empirical specifications that follow.

Table 1 also contains some information about the levels of exposure of OGD in these communities. The intensity of drilling for those very close to development is much greater within 5 km than for those a little farther away: those living 1 km from a well are near 126 wells, compared with 45 wells for those within 1–5 km. The average number of wells drilled within 5 km in the year of birth is fairly similar, however, indicating that I am comparing similar communities with respect to drilling intensity, just exploiting proximity (close versus farther away) to identify the causal impact of interest.

Online Appendix Table A2 provides summary statistics for the universe of births in Colorado from 2000 to 2011. The first column provides information on all births, and the second column provides information on births to mothers residing within 5 km of where an oil or gas well has been drilled or will eventually be drilled. The localized data I use in this analysis is actually quite similar to the characteristics of the rest of the state. Column 3 provides a decomposition of birth weight of residences within 2 km of a well to gauge the importance of the various observable mother characteristics. These control

8 In the analysis that follows, the sample is restricted to those mothers' residences within 5 km of an oil or gas well that was drilled within two years (before and after) of the timing of birth to limit other confounding and unobserved changes in these communities. I test robustness of this restriction in Section VI.D.

TABLE 1. Summary statistics for difference-in-differences sample

	Sample means		T-statistic
	Within 1 km	1–5 km	of difference
Characteristics of birth			
Birth weight (grams)	3,299.98	3,267.58	−4.09[a]
Gestation length (weeks)	38.86	38.80	−2.13[c]
Preterm birth (PTB)	0.065	0.074	2.43[c]
Low birth weight (LBW)	0.055	0.060	1.30
Very low birth weight (vLBW)	0.007	0.009	1.73
Female child	0.48	0.49	2.37[c]
Mother's demographic characteristics			
Age (years)	27.50	27.01	−5.38[a]
Education (years)	13.59	13.26	−5.90[a]
White	0.955	0.924	−8.53[a]
Black	0.004	0.022	9.66[a]
Asian	0.016	0.021	2.24[c]
Other race	0.025	0.034	3.46[a]
Hispanic	0.306	0.347	5.77[a]
Smoked during pregnancy	0.067	0.089	5.45[a]
Married	0.770	0.736	−5.14[a]
Previous risky pregnancy	0.204	0.207	0.48
Parity	2.082	2.049	−1.91
Characteristics of oil and gas wells near residence			
Average distance to closest well (km)	0.61	2.55	137.90[a]
Number of wells drilled within 5 km in year of birth	5.21	6.52	11.00[a]
Number of wells drilled within 5 km	126.48	44.74	−74.70[a]
Sample size	6,448	14,241	

Note: [a]$p < 0.01$, [c]$p < 0.10$.

variables are included in all my subsequent regression analyses, but for simplicity, I do not report these coefficients in the main tables of the paper.

IV. Graphical Evidence

If living close to a drilled well has a negative impact on infant health at birth, we should see average birth weight fall after drilling begins for mothers' residences in close proximity to wells. We should also observe larger impacts for homes closest to drilling activity. Most important, the causal effect is supported if the reduction in birth weight (gestation) close

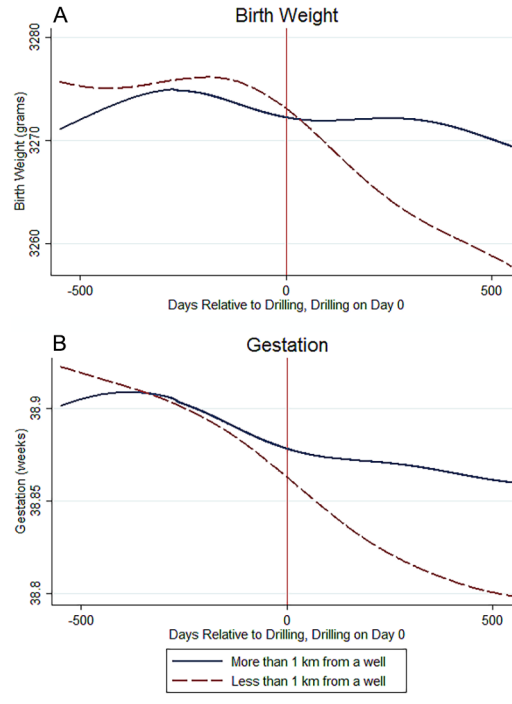

FIGURE 1. Birth weight and gestation trends before and after drilling within 5 km. A color version of this figure is available online.

to a well coincides with when drilling begins and does not reflect a preexisting downward trend in health. Figure 1 shows the birth weight and gestation gradient of time with respect to when drilling begins within 1 km and 1–5 km of the well locations. This gradient is measured for births 500 days before and during/after drilling. If the birth weight decline reflected a preexisting trend, we would see a consistent downward trend over this time period. Instead, I find a fairly sharp decrease in birth weights (gestation) coincident with the spud dates (defined as time = 0).[9] Figure 1 also provides supportive evidence that there are parallel trends in the outcomes prior to drilling.

V. Empirical Strategy

In order to estimate a causal estimate of the effect of OGD on reproductive health outcomes in affected communities, I employ a difference-in-differences research design. In this

9 The spud date is the date that the drill bit hit the ground, but building the well pad happens about one to three months prior to that date on average. These graphs also use birth date as the relevant date, when it is unlikely that drilling happening on the child's birth date (e.g., time = 0 in the graph) would influence birth outcomes. Because of the somewhat fuzzy nature of this timing, there does appear to be a slight trend prior to time = 0, but for the most part, these graphs suggest that the differences coincide with the timing of drilling.

paper, my affected or "treated" group are those babies who are born after a well is drilled within 1 km of the mother's residence. My comparison groups are those mothers who live 1 km from wells that have not yet been drilled or those mothers who live 1–5 km from wells, before or during/after drilling. This approach allows me to exploit the expected dose-response relationship: the closer a mother lives to a well, the more likely she is exposed, and the stronger the potential detected effect. But it also has the advantage of comparing mothers who all live relatively close (all within 5 km or less than 3 miles) to development and therefore are likely to be more homogeneous in both observable and unobservable factors.

An important caveat for my empirical strategy is that I can observe health at birth for only those babies that are born alive. Also, I can only observe births for those mothers who choose to get pregnant or can get pregnant. If the composition of mothers choosing to get pregnant changes with the introduction of resource extraction, then the health that I observe may not be indicative of the average health of those living near wells in these neighborhoods. Furthermore, OGD has been shown to increase migration, and this factor could also change the composition of mothers living near drilling (Wilson 2022).[10] There are a few ways that I address this issue. First, I show that most observable characteristics are not changing in response to development activities.[11] While this does not mean that unobservables are unchanged, it provides some reassurance that this identification strategy is sound. Second, for mothers who have more than one baby during my window of interest, I study whether they move after drilling began and do not find any statistically significant changes in mobility in response to development activities.[12] Third, although I cannot measure unobserved miscarriages, I test whether there is a change in sex ratio in these communities in response to development activities and find that there are no changes in sex ratio. This indicates that my findings are not likely to be driven by selection into a live birth. Finally, owing to concerns about historical drilling contaminating the results and migration driving the results, I limit the analysis to births that occur within two years of the nearest well's spud date (i.e., birth date +/−2 years of spud date) for the main results.[13]

10 It is possible that mothers who value their children's health are more likely to move away from communities where drilling is taking place. This migratory effect would lower the average health of the observed births. However, it is also possible that those who are more likely to move are families who are experiencing the worse health effects of drilling. This migratory effect would increase the average health of the observed births. Thus, it is not clear whether selection or composition of the mother characteristics would lead me to overestimate or underestimate the health impacts of close proximity to drilling activity.

11 There is an increase in teenage mothers after drilling, which reflects that areas with drilling had fewer teen mothers prior to drilling than did areas farther from drilling. See Table 1 and Online Appendix Figure A2 for distance gradients of maternal characteristics. I control for maternal age in my main results and also run regressions excluding teen mothers, and show that my main results are not driven by this change.

12 There is no statistically significant relationship between the difference-in-differences estimator and the probability of moving; however, some magnitudes could be interpreted as economically meaningful (Online Appendix Table A18). To address this issue, I also estimate regressions on mothers who do not move, and the results are qualitatively similar to my main results (Online Appendix Table A19).

13 I show robustness to this decision in Online Appendix Tables A14 and A15 and show no restriction, within four years (i.e., +/−4 years) and within two years. This issue is discussed in Section VI.D.

My main equation takes the following form:

$$Outcome_{ijt} = \beta_0 + \beta_1 D_{ijt}^{1km} + \beta_2 Post_{ijt} + \beta_3 D_{ijt}^{1km} * Post_{ijt} + \beta_4 X_{ijt} + \gamma_t + \omega_j + \epsilon_{ijt} \quad (1),$$

where $Outcome_{ijt}$ is birth weight, gestation, premature birth, or low birth weight for each infant i born in location j in quarter-year t. $Post_{ijt}$ is an indicator for the birth occurring after the spud date of wells within 10 km, and D_{ijt}^{1km} is an indicator for whether the residence was 1 km from a current or future well. B_3 is the coefficient of interest on $D_{ijt}^{1km} * Post_{ijt}$ and is the difference-in-differences estimator (i.e., an indicator for wells drilled prior to birth within 1 km). The term ω_j is a zip code fixed effect and is designed to capture any unobserved time-invariant characteristics of each zip code in the sample.[14] γ_t quarter-by-year fixed effects are included to control for systematic trends over time.[15] The standard errors in these models are clustered at the mother's residence zip code. The vector X_{ijt} contains mother and child characteristics: indicators for whether the mother is white, black, Asian, other race (left out category), and/or Hispanic; four mother education categories (less than high school [left out category], high school, some college, and college or more); mother age categories (teen [left out category], 19–24, 25–34, and 35+); indicators for smoking or drinking during pregnancy; mother's marital status; previous risky pregnancy; current risky pregnancy; parity; and an indicator for sex of the child. Starting in 2007, the birth certificate record contained information on insurance payment type (e.g., Medicaid, private insurance, self-pay, other), receipt of support from the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), and income categories.[16]

I augment the main model in multiple ways: (1) mother fixed effects,[17] (2) nearest well fixed effects,[18] and (3) interaction of the post dummy with the number of wells for the intensive margin (i.e., density of drilling). I also change the comparison group to be within 2 km (shown in Online Appendix Table A13) and within 15 km and no distance restriction (shown in Online Appendix Tables A14 and A15).

VI. Results

A. DIFFERENCES IN CHARACTERISTICS OF MOTHERS RESIDING CLOSE TO A WELL

My estimation strategy hinges on the relative similarity between mothers residing within 1 km of a well and mothers residing within 1–5 km of a well at the time of the observed

14 I also report models where j is (1) county in Online Appendix Table A11; (2) well ID for nearest well in Online Appendix Table A5; and (3) mother ID in Online Appendix Table A4. Zip code is the smallest geographical unit that allows for studying the same communities over time, given that the nearest well can change with more drilling and mothers who have more than one birth may be different from those who have one child. This is why the zip code fixed-effects models are my preferred specification.

15 I show that the main results are robust to month-by-year fixed effects in Online Appendix Table A12.

16 See Online Appendix Table A16.

17 Standard errors are clustered at the county.

18 Standard errors are clustered at the well.

TABLE 2. Differences in average characteristics close to well locations

	Teen mom (1)	High school dropout (2)	Hispanic (3)	Smoked (4)	Married (5)	Risky pregnancy (6)
	A. Pre-drilling differences in characteristics					
Within 1 km of well	−0.0260[a]	−0.0513[b]	−0.0446	−0.0106	0.0369	−0.0076
	(0.0055)	(0.0253)	(0.0341)	(0.0080)	(0.0247)	(0.0116)
Sample size	15,705	15,705	15,705	15,705	15,705	15,705
R^2	0.028	0.133	0.190	0.044	0.046	0.200
	B. Pre- and post-drilling differences in characteristics					
Post-drilling	0.0231[b]	−0.0086	−0.0088	−0.0321	−0.0063	−0.0525[a]
	(0.0096)	(0.0213)	(0.0241)	(0.0210)	(0.0297)	(0.0191)
Within 1 km	−0.0319[a]	−0.0475[c]	−0.0461	−0.0083	0.0390	−0.0031
	(0.0058)	(0.0257)	(0.0355)	(0.0087)	(0.0265)	(0.0102)
Within 1 km × post-drilling	0.0262[a]	0.0227	0.0045	0.0017	−0.0140	−0.0116
	(0.0046)	(0.0182)	(0.0243)	(0.0072)	(0.0176)	(0.0085)
Sample size	20,689	20,689	20,689	20,689	20,689	20,689
R^2	0.029	0.124	0.176	0.041	0.044	0.208

Note: Each coefficient is from a different regression. Pre-drilling (post-drilling) refers to births that occur before (after) the spud date of the closest well. Standard errors are clustered at the mother's residence zip code. The sample is limited to residences within 5 km of a well, within two years of the spud date, and singleton births. All regressions include residence zip code and quarter × year fixed effects. [a]$p < 0.01$, [b]$p < 0.05$, [c]$p < 0.10$.

birth. While this is somewhat supported by the graphical evidence, I formally estimate these differences. I proceed by estimating the pre-drilling differences for those within 1 km using the full sample of residences within 5 km.

In Table 2, panel A, I limit the sample to births that took place before the drilling began. I find little evidence of any preexisting differences in either health at birth or mother characteristics that would be indicative of worse health trends in these communities prior to drilling. Although some differences are statistically significant, these communities boast heavier babies, more education, and fewer teen mothers prior to drilling within 1 km of a future well. All of these characteristics indicate potentially better health outcomes, not worse. This does suggest selection into the locations where drilling takes place, but it goes in a direction that we may not have anticipated. Often, new industrial activity is correlated with poor, less educated communities. Fortunately, any bias would push my main estimates towards a null effect, and any adverse effects detected are certainly not reflective of a preexisting adverse health trend as measured by maternal characteristics.

To further test the validity of my research design, I also estimate equation 1 and use the difference-in-differences estimator to determine whether there are any changes in mother characteristics after drilling began. Results are reported in Table 2, panel B. Mothers who gave birth after drilling occurred within 1 km of their residence have an increased likelihood to be teen mothers, consistent with at least one other study that found an increase of 7.2 births per 1,000 among teenage white mothers (Owen 2022). Although this shows a change in the composition of mothers, teen mothers still make up a very small proportion of the population (~6 percent; see Online Appendix Table A2). I also estimate models removing teen mothers and show that the results are consistent and that this change in the composition of mothers is not driving the primary effect (see Online Appendix Table A8, panel E). In my main models discussed in the next section, I include controls for these and more characteristics in all specifications to help account for any changes in the composition of who selects into a live birth after drilling begins.

B. THE IMPACT OF OIL AND GAS EXTRACTION ON INFANT HEALTH

I present my main findings in Table 3, with and without maternal characteristics. Estimates with and without characteristics do not change much in magnitude or significance, providing confidence in the estimation strategy. Estimating equation 1 in Table 3—my difference-in-differences specification—I find that birth outcomes are adversely affected by drilling activity within 1 km of the maternal residence. The estimate for birth weight is 35.79 grams, with mother characteristics included, and is statistically significant at the 1 percent level. In addition to reduced average birth weight, residences within 1 km of a well have births with reduced gestational periods, increased prevalence of LBW of 1.70 percentage points, and increased prevalence of premature birth of 2.15 percentage points.[19] To explore whether this effect is persistent, I present the estimates for distance bins out to 5 km in Figure 2 and present in Online Appendix Table A3 estimates from equation 1 using 1.5 km; I find a persistent, but less precise, reduction of birth weight of 25.55 grams, on average. This effect does not persist past 2 km and suggests that there may be a threshold of exposure between 1.5 km and 2 km from an oil or natural gas well (approximately 1 mile).

To assess the robustness of these results, I also estimate equation 1 with mother fixed effects and well fixed effects (Online Appendix Tables A4 and A5). Comparing siblings, except for gestation and premature birth, however, the qualitative magnitudes are similar to my main results and are statistically significant (Table A4).[20] Comparing births around the same nearest well is another possible way to identify the effects of drilling on birth outcomes. Unfortunately, the nearest well over time is not a constant spatial location, with new wells being drilled across the landscape. Furthermore, the average residence that has a well at any time within 1 km of the maternal residence is exposed to an average of

19 Because of the imprecise nature of the oil and gas well dates, I also estimate the main results using conception date as the date of interest to define exposure. These results are qualitatively similar to the main findings and are available upon request.

20 Gestation and prematurity are most often predicted by prior gestation or prematurity within the same mom (American College of Obstetricians and Gynecologists 2021), which may explain why these outcomes are not as robust.

TABLE 3. Impact of well location on birth outcomes

	Birth weight		Gestation		Low birth weight		Preterm birth	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. Pre-drilling differences in birth outcomes							
Within 1 km of well	67.0400[a]	52.2900[a]	0.1250[a]	0.1270[a]	−0.0147[c]	−0.0114	−0.0264[a]	−0.0252[a]
	(10.9100)	(11.2400)	(0.0401)	(0.0354)	(0.0076)	(0.0074)	(0.0050)	(0.0049)
Sample size	15,703	15,703	15,704	15,704	15,703	15,703	15,704	15,704
R^2	0.017	0.055	0.016	0.029	0.013	0.028	0.012	0.022
Maternal characteristics	No	Yes	No	Yes	No	Yes	No	Yes
	B. Pre- and post-drilling in birth outcomes							
Post-drilling	−0.1150	−7.9090	−0.0431	−0.0695	0.0057	0.0096	−0.0063	−0.0023
	(30.1300)	(28.7700)	(0.0607)	(0.0647)	(0.0118)	(0.0121)	(0.0082)	(0.0080)
Within 1 km	61.7900[a]	48.4700[a]	0.1280[a]	0.1290[a]	−0.0159[b]	−0.0130[c]	−0.0250[a]	−0.0236[a]
	(11.1400)	(10.8700)	(0.0441)	(0.0401)	(0.0068)	(0.0066)	(0.0046)	(0.0045)
Within 1 km × post-drilling	−38.8900[a]	−35.7900[a]	−0.1080[b]	−0.1140[a]	0.0180[a]	0.0170[a]	0.0216[a]	0.0215[a]
	(11.4000)	(11.1300)	(0.0421)	(0.0408)	(0.0056)	(0.0054)	(0.0044)	(0.0046)
Sample size	20,687	20,687	20,687	20,687	20,687	20,687	20,687	20,687
R^2	0.014	0.055	0.014	0.025	0.011	0.025	0.011	0.021
Maternal characteristics	No	Yes	No	Yes	No	Yes	No	Yes

Note: Each coefficient is from a different regression. Post-drilling refers to births that occur after the spud date of the closest well. Standard errors are clustered at the mother's residence zip code. The sample is limited to residences within 5 km of a well, within two years of nearest spud date, and singleton births. All regressions include residence zip code and quarter × year fixed effects. Maternal characteristics include race (black, Hispanic, Asian), education (high school, some college, college), age (19–24, 25–34, 35+), female child, smoking during pregnancy, drinking during pregnancy, indicators for parity, an indicator for previous/current risky birth, and marital status. [a]$p < 0.01$, [b]$p < 0.05$, [c]$p < 0.10$.

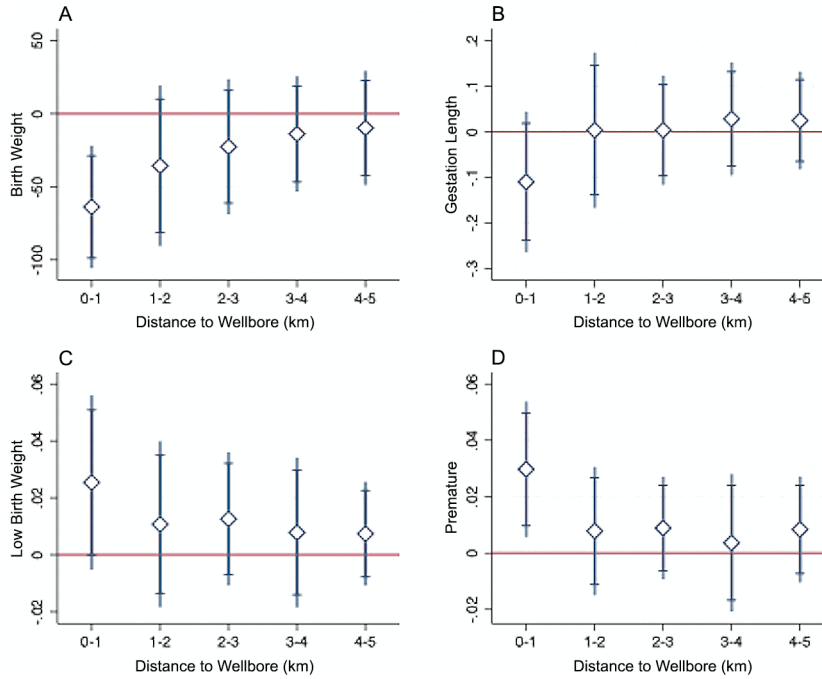

FIGURE 2. Birth outcome impacts of drilling within 5 km. Figure plots coefficients from a model with binary distance bins (0–1, 1–2, . . . , 4–5) indicating any drilling before birth within that bin. Also included are binaries for wells within these distance bins ever and a binary for post-drilling. The model is limited to the main sample described in Table 3 and includes the same controls and fixed effects. A color version of this figure is available online.

seven wells within 1 km prior to birth.[21] Despite these limitations, I show in Table A5 that my main results are robust to including the nearest well as a fixed effect.

B.1. INTENSIVE MARGIN: WELL DENSITY. In Table 4, I present results analogous to Table 3, panel A, except with a continuous measure of treatment defined by number of wells drilled prior to birth. In Table 4, panel B, I show the impact of well density by when wells were drilled, either prior to conception or during gestation (i.e., in utero). Here, I present the intensive-margin results with zip code fixed effects (as in Table 3) and mother fixed effects (as in Online Appendix Table A4). The intensive-margin results consistently suggest an adverse health impact associated with each additional well drilled before birth

21 If I drew concentric circles around wells to identify births within some proximity of a central well, many times these births would be assigned to multiple wells and I would need to weight my regression appropriately. This led me to compare within zip codes, which are the smallest unit to capture communities over time. It is also why I report models using the density of wells, which does not easily translate to a well fixed-effects model.

TABLE 4. Impact of well density on birth outcomes

	Birth weight		Gestation		Low birth weight		Preterm birth	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. No. of wells before birth							
No. of wells within 1 km	1.4250[b]	−0.7130	0.0047[b]	−0.0221[a]	−0.0007[b]	0.0010[c]	−0.0004	0.0004
	(0.6480)	(1.1590)	(0.0021)	(0.0036)	(0.0003)	(0.0005)	(0.0003)	(0.0011)
Post-drilling	−7.9620	−0.0392	−0.0787	0.0050	0.0087	−0.0300	−0.0024	−0.0186
	(29.6400)	(53.6400)	(0.0661)	(0.1210)	(0.0126)	(0.0235)	(0.0083)	(0.0170)
No. of wells within 1 km × post-drilling	−1.1700	−1.8210	−0.0034	−0.0009	0.0008[b]	0.0028[c]	−0.0004	−1.78e−05
	(0.9820)	(2.0650)	(0.0032)	(0.0046)	(0.0004)	(0.0015)	(0.0003)	(0.0007)
	B. No. of wells before conception and in utero							
No. of wells within 1 km	1.4010[b]	−0.6820	0.0045[b]	−0.0222[a]	−0.0007[b]	0.0010[c]	−0.0004	0.0004
	(0.6540)	(1.0980)	(0.0021)	(0.0036)	(0.0003)	(0.0005)	(0.0003)	(0.0011)

Post-drilling	−8.6160	1.9790	−0.0856	0.0118	0.0089	−0.0306	−0.0022	−0.0191
	(29.9100)	(54.0100)	(0.0661)	(0.1250)	(0.0126)	(0.0245)	(0.0083)	(0.0166)
No. of wells within 1 km × before conception	−1.4250	−2.0170	−0.0053	−0.0046	0.0006[c]	0.0032[b]	−0.0003	0.0003
	(0.8760)	(1.7630)	(0.0046)	(0.0044)	(0.0003)	(0.0015)	(0.0004)	(0.0009)
No. of wells within 1 km × in utero	0.0156	−2.1290	0.0061	0.0060	0.0015[b]	0.0021[c]	−0.0005	−0.0007
	(2.1230)	(1.8650)	(0.00451)	(0.0148)	(0.00065)	(0.0012)	(0.0004)	(0.0006)
Sample size	20,653	4,062	20,653	4,060	20,653	4,062	20,653	4,060
R^2	0.053	0.706	0.025	0.634	0.024	0.559	0.020	0.564
Dep. var. mean	3,278	3,298	38.82	38.80	0.0582	0.0544	0.0714	0.0663
Zip code fixed effects	Yes		Yes		Yes		Yes	
Mother fixed effects		Yes		Yes		Yes		Yes

Note: Each column and panel represents a separate regression. The sample is limited to residences within 5 km of a well and singleton births. All regressions include quarter × year fixed effects. Maternal characteristics include race (black, Hispanic, Asian), education (high school, some college, college), age (19–24, 25–34, 35+), female child, smoking during pregnancy, drinking during pregnancy, indicators for parity, an indicator for previous/current risky birth, and marital status. Standard errors are clustered at the mother's residence zip code for zip code fixed effects and county for mother fixed effects. [a]$p < 0.01$, [b]$p < 0.05$, [c]$p < 0.10$.

for birth weight, gestation, and low birth weight (LBW), although these results are only statistically significant for LBW. The results for premature birth are opposite-signed and not precisely estimated. Unlike the extensive margin in a mother fixed-effects model (Online Appendix Table A4), I find a statistically significant effect only for low birth weight. It is possible that I do not have enough sample to estimate the impact of the intensive margin (i.e., each additional well drilled) across siblings. This finding is consistent with those from Pennsylvania: Hill and Ma (2022) found strong impacts of the intensive margin for exposure to gas wells near public drinking water sources in a mother fixed-effects model with a larger sample exposed, but Currie, Greenstone, and Meckel (2017) did not find statistically precise effects of the extensive margin when assessing proximity to maternal residence, because of a small sample. In panel B of Table 4, I find suggestive evidence that wells drilled before conception as well as wells drilled during pregnancy have impacts on adverse birth outcomes. In Online Appendix Figure A3, I show coefficients for number of wells in different distance bins and find that the majority of the impact on birth weight is within 1 km, consistent with my extensive-margin results (see Online Appendix Table A6 for full specification coefficients).[22]

B.2. LEVELS OF LOW BIRTH WEIGHT AND PREMATURE BIRTH. Low birth weight and premature birth are arguably arbitrary cut-offs and are an attempt to get at a latent variable—the threshold at which children require additional medical support in their first few weeks of life. Online Appendix Table A7 reports three categories of severity for each of these measures. For levels of low birth weight, oil and gas extraction primarily increased the probability of being slightly low birth weight by 1.1 percentage points and increased the probability of being very low birth weight by 0.3 percentage points. In contrast, exposure increased the rate of all levels of premature birth: the probability of being slightly premature (34–37 weeks) increased by 1.3 percentage points, the probability of being moderately premature (32–34 weeks) increased by 0.3 percentage points, and the probability of being very premature increased by 0.5 percentage points.

B.3. HETEROGENEITY BY MATERNAL CHARACTERISTICS. Birth outcomes are associated with different subgroups, as shown in the marginal effects reported in Online Appendix Table A2. Prior work has shown that families of lower socioeconomic status are more likely to be impacted by environmental exposures (Almond, Currie, and Duque 2018). In Online Appendix Table A8, I present results for various potentially important subgroups. Most notably, the results for white non-Hispanic mothers are quite a bit more pronounced than the main results in the paper. In addition, the results for smokers are very large, suggesting that maternal smoking may have a compounding effect.[23] Some may be concerned that migration is driving these results. Although it is an imperfect measure of mobility, I find very similar results for mothers who were born in Colorado. Lastly, as previously discussed, there are more teen mothers after drilling began, which could be driving

22 The coefficients change slightly when the number of wells in other distance bins are included in the model. Table 4 includes only number of wells within 1 km, whereas Online Appendix Table A6 controls for number of wells within each distance, as shown in the table. The specification where multiple distance bins are included results in the effect on birth weight becoming slightly larger and statistically significant.

23 This result has been found in other papers as well (Currie, Neidell, and Schmieder 2009).

my results. In Table A8, panel E, I present the results limited to mothers aged 19–35 years old, and the main results are robust to this restriction. Therefore, I conclude that migration or changes in composition may influence my main results but are not driving them entirely.

B.4. HETEROGENEITY BY WELL TYPE AND PRODUCTION. This paper is motivated by a dearth of research using quasi-experimental approaches to assess the health impacts of all drilling types including oil, natural gas, coalbed methane, and vertical/horizontal drilling.[24] Thus far I have been pooling all well types and also not accounting for the production status of wells nearby at the time of pregnancy and birth. In Table 5, I present heterogeneity of impacts by producing wells (panel A), horizontal wells (panel B), and resource type (panel C). I find that most of the impacts in my main results are from newly spudded wells, but that producing wells may also add some adverse effect that is not statistically significant (Table 5, panel A). For horizontal wells, I find that they are not very different from vertical wells in terms of impacts (Table 5, panel B). This finding is consistent with other studies; however, most other research finds a slightly larger impact for horizontal wells (Willis et al. 2021). Lastly, it appears that coalbed methane wells are contributing the most to adverse birth outcomes, but that oil and natural gas wells (if you add the post-drilling and oil or gas interaction terms) are still contributing to the impacts I estimate in the main models. In particular, coalbed methane wells seem to have a very large impact on preterm birth (Table 5, panel C, column 4).

C. IMPACT OF DRILLING ON MATERNAL HEALTH AND HEALTH CARE

I find some indication in this paper of positive selection into living near drilling in Colorado: families who live very close (within 1 km) to drilling have higher income and are more highly educated (See Section VI.A). In Online Appendix Table A9, I find some indication of pre-drilling differences that could indicate better health and resources with respect to reproductive health for families living within 1 km of future drilling. For example, mothers who gave birth within 1 km from future drilling were 2.6 percentage points less likely to have labor complications and 2.8 percentage points more likely to have adequate prenatal care compared with those who live 1–5 km away from future drilling.

A growing literature has looked at maternal health to identify stress during pregnancy (Camacho 2008; Torche 2011; Currie and Rossin-Slater 2013; Currie and Schwandt 2014). This literature uses hypertension, eclampsia, and diabetes as medical risk factors correlated with maternal stress (Bussières et al. 2015). Hypertension, eclampsia, and gestational diabetes have also been linked to air pollution exposure during pregnancy in the epidemiology literature (Pereira et al. 2013; van den Hooven et al. 2011; Woodruff, Morello-Frosch, and Jesdale 2008) and exposure to unconventional drilling in Texas (Willis et al. 2022). I label these three conditions as complications of pregnancy and combine them into one indicator. Table 6 presents results using the difference-in-differences estimator to look at maternal health factors. Interestingly, living within 1 km of an oil or gas well increases the probability of complications of pregnancy (defined as either hypertension, eclampsia,

24 Notable exceptions are two studies in Texas on birth outcomes (Willis et al. 2021) and childhood asthma (Willis et al. 2020).

TABLE 5. Heterogeneity of impacts by well type

	Birth weight (1)	Gestation (2)	Low birth weight (3)	Preterm birth (4)
	A. Producing wells			
Within 1 km	36.7400[a]	0.0742[b]	−0.0180[a]	−0.0063
	(11.2400)	(0.0371)	(0.0054)	(0.0070)
Post-drilling	−8.0210	−0.0701	−0.0023	0.0095
	(28.7300)	(0.0645)	(0.0080)	(0.0123)
Within 1 km × post-drilling	−19.5700[c]	−0.0315	0.0151[b]	0.0109
	(10.3600)	(0.0341)	(0.0062)	(0.0067)
Within 1 km × post-drilling × producing	−7.7240	−0.0736	−0.0056	−0.0174[b]
	(22.2900)	(0.0613)	(0.0116)	(0.0072)
	B. Horizontal wells			
Within 1 km	36.2000[a]	0.0708[c]	−0.0173[a]	−0.0057
	(11.3100)	(0.0361)	(0.0051)	(0.0066)
Post-drilling	−7.7480	−0.0687	−0.0028	0.0089
	(28.7400)	(0.0648)	(0.0080)	(0.0124)
Within 1 km × post-drilling	−21.7900[c]	−0.0446	0.0185[b]	0.0142
	(12.6400)	(0.0477)	(0.0074)	(0.0087)
Within 1 km × post-drilling × horizontal	3.1820	0.0120	−0.0090	−0.0116
	(15.7200)	(0.0566)	(0.0061)	(0.0078)
	C. Oil, gas, or coalbed methane wells			
Within 1 km	35.1900[a]	0.0748[b]	−0.0184[a]	−0.0059
	(10.9100)	(0.0373)	(0.0053)	(0.0069)
Post-drilling	−6.4050	−0.0729	−0.0020	0.0086
	(28.3600)	(0.0645)	(0.0080)	(0.0120)
Within 1 km × post-drilling	−63.4900[c]	0.0323	0.0124	0.0344[a]
	(34.0100)	(0.0966)	(0.0131)	(0.0111)
Within 1 km × post-drilling × oil	44.6300	−0.0713	0.0016	−0.0257[b]
	(36.9400)	(0.0924)	(0.0124)	(0.0107)
Within 1 km × post-drilling × gas	25.3700	−0.0841	0.0117	−0.0229[b]
	(20.0600)	(0.0631)	(0.0085)	(0.0092)

TABLE 5. *Continued*

	Birth weight (1)	Gestation (2)	Low birth weight (3)	Preterm birth (4)
Sample size	20,653	20,653	20,653	20,653
R^2	0.053	0.024	0.020	0.024
Zip code fixed effects	Yes	Yes	Yes	Yes
Distance restriction (km)	5	5	5	5
Dep. var. mean	3,278	38.82	0.0714	0.0582

Note: Each column represents a separate regression. Standard errors are clustered at zip code. The sample is limited to a four-year window surrounding drilling (i.e., two years prior and two years after), residences within 5 km of a well, and singleton births. All regressions include mother residential zip code, and quarter × year fixed effects. Maternal characteristics are included (see Table 3 for list). $^{a}p < 0.01$, $^{b}p < 0.05$, $^{c}p < 0.10$.

or gestational diabetes) by 1.1 percentage points (~6 percent). This finding is consistent with a study in Texas that found an increase in maternal hypertension (5 percent) and eclampsia (26 percent) in communities with OGD (Willis et al. 2022).

In Table 6, I do not find increased prevalence of lung disease, but maternal asthma is not well reported in these data and I do not find any changes in prenatal care of the mothers associated with oil and gas extraction, which indicates that the adverse birth and maternal outcomes are not caused by changes in access to or utilization of prenatal care by mothers exposed to drilling. The male/female sex ratio has been used to measure selection into live birth (Sanders and Stoecker 2015), and here I find little evidence of oil and gas extraction increasing the prevalence of miscarriage, indicating that the adverse birth outcomes observed are unlikely to be related to selection into live birth. Finally, I do not find strong evidence that fertility increased in counties with residences within 1 km of drilling (less than 1 percent increase). This result is not consistent with other research that has found large increases in birth rates at the commuting zone level associated with the shale boom (Kearney and Wilson 2018). I include in Online Appendix Table A10 additional ways to measure changes in fertility.[25] Using the intensive margin, I find that drilling within counties that have wells within 5 km may have less fertility (less than 1 percent change per additional horizontal or producing well) for contemporaneous drilling. This leads me to

25 I aggregate to counties because births in zip codes are sparse for this analysis: the 25th percentile was 4 births per year and the median was 24. In panels A–C of Online Appendix Table A10, I explore various definitions of exposure defined at the county level. The migration story for jobs may respond to both contemporaneous drilling and cumulative drilling. Panel A explores number of wells by various types (horizontal, vertical, producing) drilled in the year. Panel B looks at binary exposure (any wells drilled in the year). Perhaps we think it is instead cumulative drilling that could impact fertility, and so I include panel C for cumulative wells by type. Given that my study is at a more micro level than county, I retained in panel D the exposure to drilling at the maternal residence for births within each county to address issues of drilling occurring within counties but not near maternal residences.

TABLE 6. Impact of well location on maternal health, prenatal care, and fertility

	Complications of pregnancy (1)	Lung disease (2)	Labor complications (3)	Fetal death (male/female) (4)	Prenatal (5)	ln(births) (6)
Post-drilling	0.0060	0.0237[a]	−0.0462[b]	−0.0281	−0.1040[a]	−0.0259
	(0.0041)	(0.0039)	(0.0188)	(0.0271)	(0.0100)	(0.0343)
Within 1 km	−0.0076	0.0003	−0.0006	−0.0030	0.0404[a]	0.0401
	(0.0070)	(0.0024)	(0.0089)	(0.0091)	(0.0113)	(0.0420)
Within 1 km × post-drilling	0.0108[b]	−0.0039	0.0077	−0.0164[b]	0.0090	0.0096
	(0.0044)	(0.0028)	(0.0162)	(0.0071)	(0.0094)	(0.0417)
Sample size	20,689	20,689	20,689	20,689	20,576	450
R^2	0.395	0.723	0.128	0.039	0.060	0.995

Note: Each coefficient is from a different regression. Pre-drilling (post-drilling) refers to births that occur before (after) the spud date of the closest well. Standard errors are clustered at the mother's residence zip code. The sample is limited to residences within 5 km of a well and singleton births. All regressions include residence zip code and quarter × year fixed effects. Pre- and post-drilling regressions also include maternal characteristics. See Table 3 for covariates included. Column 6 is a county-level model. [a]$p < 0.01$, [b]$p < 0.05$.

conclude that fertility may be changing in response to drilling but is not likely large enough to be driving my results.

D. ROBUSTNESS CHECKS AND MOBILITY

To provide additional support for the research design, I perform a few robustness checks. I assess robustness to spatial fixed effects, distance restrictions, temporal restrictions, and covariate inclusions. I also estimate placebo regressions. Finally, I assess the main threat to identification: mobility. I measure the likelihood that a mother moves across pregnancies, and I restrict the analysis to the sample that does not move. My main results are qualitatively robust to these checks.

D.1. FIXED EFFECTS, SPATIAL AND TEMPORAL RESTRICTIONS. Most of the results reported in this paper use zip code fixed effects. I have also estimated all of the results using county fixed effects that also include county by year by month-of-birth fixed effects. A comparison with the main results is reported in Online Appendix Table A11. Choice of location fixed effects does not appreciably change the results. I also test whether quarter by year fixed effects are sufficient to capture seasonal trends in the outcomes by estimating my main results with month-by-year fixed effects. The results are robust to the inclusion of month-by-year fixed effects in Online Appendix Table A12.[26]

26 I also show robustness to inclusion of month-by-year fixed effects in Online Appendix Table A4 for the mother fixed-effects specifications, and the results are qualitatively similar for birth weight and low birth weight.

In my main specification, I restrict the analysis to within two years of a spud date and within 5 km of the nearest well. In Online Appendix Table A13, I show robustness to comparing within 1 km and 1–2 km. In Online Appendix Tables A14 and A15, I present regressions relaxing these restrictions for continuous outcomes (birth weight and gestation) and binary outcomes (LBW and PTB), respectively. In columns 1–3, I show no spatial restriction (i.e., all of Colorado), and show no temporal restriction, then two-year and four-year restrictions. Analogously, I do the same in columns 4–6 for within 5 km, and in columns 7–9 for within 15 km, varying the temporal restriction in each column. I find that the results are strongly robust for the continuous outcomes (Table A14) in both magnitude and statistical significance, but are less robust for the binary outcomes (Table A15; sign and magnitude, but not statistically precise). Across the board, results are stronger with some temporal restriction, but are robust across any spatial restrictions (holding the temporal restriction constant). This finding further provides suggestive evidence that the main results reflect impacts from newer drilling and not from older producing wells. It is also consistent with Hill and Ma (2022), who found drinking water impacts during gestation but not in models that measured cumulative drilling that occurred prior to pregnancy. I, however, find impacts for pre-conception drilling in Table 4, for drilling that occurs within two years of birth.

D.2. ADDITIONAL INCOME AND INSURANCE COVARIATES. In Online Appendix Table A16, I present difference-in-differences estimates for the four birth outcomes for the years 2007–11. The birth certificate records changed in 2007 to include the mother's income, insurance status, and whether she received public support from the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC). To make sure that the results are robust to additional controls, especially controls for socioeconomic status like income and insurance, I replicate the main table using this subsample. For each health measure, I provide two columns. The first column includes the same controls as the rest of the paper, and the second column adds these additional income/insurance controls. The results suggest that a drilled well within 1 km of a mother's residence from 2007 to 2011 results in a decreased birth weight of 30.7 grams and a reduced gestational period of 0.10 weeks, on average. The estimates also suggest a 2.03 percentage point increase in low birth weight and a 1.16 percentage point increase in premature birth. Only the premature estimate becomes less precise with the inclusion of expanded controls for income. These estimates also suggest that the main results in the paper are not driven by earlier time periods, and that the impacts of close proximity to oil and gas wells remain persistent in more recent years.

D.3. PLACEBO REGRESSION. Owing to my analysis showing evidence of preexisting differences in the communities located closest to drilled wells relative to communities close to future wells or those who live a little farther away, I employ a few placebo regressions. Even though the trends are suggestive of better outcomes prior to drilling, it is theoretically possible that the increase in adverse birth outcomes after drilling is driven by differential trends in fertility or migration post-drilling. I investigate this possibility by estimating equation 1 using false spud dates two years prior to the actual spud date to define exposure. Online Appendix Table A17 presents baseline estimates and the results of this placebo regression, and I find no evidence of a spurious effect.

Using a date that is a fixed number of years prior may not be a suitable choice if different communities systematically are exposed at different times. To address this issue, I also perform checks using two random dates—one relative to the spud date and one relative to the birth date. Using a uniform distribution, I create a random date based on the number of days in the year of spud or birth and then allow the year to change randomly five years before and after. The results, presented in Online Appendix Table A17, also show no statistically significant effects and provide support for the research design used.

D.4. MOBILITY. Prior research has shown large causal impacts of OGD on migration, particularly in North Dakota (Wilson 2022). To address this threat to my identification, I look at maternal mobility, using the fact that I observe siblings in the birth certificate. I define mobility in a few ways: mother ever moved, mother moved between first and second birth, mother moved between second and third birth, and mother moved between third and fourth births. As we get to higher-order births, the sample changes because there is selection into who has more than one or two births. Taking these limitations into account, I present in Online Appendix Table A18 results for no spatial or temporal limitations and my main sample. I do not find any statistically significant effect of drilling within 1 km prior to birth on the likelihood of moving; however, the magnitudes could be economically meaningful. For example, while overall, I find a 1–4 percent increase in the likelihood of moving, in my preferred sample, I find a 7 percent and 10 percent increase in the likelihood of moving between first and second births and between second and third births, respectively. However, for mothers exposed to drilling who have four or more children, they are 19 percent less likely to move compared with mothers with four or more children farther from drilling. Considering birth order substantially changes the picture. Given these qualitative findings of possibly large mobility impacts, I also present results limiting the analysis to mothers who do not move (Online Appendix Table A19). My results are qualitatively robust to this restriction.

VII. Discussion

My study seeks to understand and quantify the impacts of oil and gas extraction on infant health. The chemicals used during drilling, cleaning drill rigs, and hydraulic fracturing are linked to birth defects, cancer, and reduced lung function, but there is little guidance from the scientific literature about the magnitude, time horizon, or likelihood of these effects. Additionally, recent studies have shown an increase in air pollution associated with drilling, but little research has been done to assess how far these air pollutants can travel.

My results suggest that babies born of mothers who lived within 1 km of an oil or gas well during pregnancy had lower birth weights on average after drilling. Oil and gas extraction increased the incidence of low birth weight and premature birth in the vicinity of a well by 31 percent and 33 percent, respectively.[27] Furthermore, birth weight and gestation were

27 Using 2007–11 data or a mother fixed effect, the results for premature birth are completely attenuated with the inclusion of maternal income, insurance type, and WIC, indicating that the effect for premature birth is less robust and likely explained by unobserved socioeconomic status or other factors.

decreased by 35.8 grams (1.1 percent) and 0.11 weeks (0.3 percent) on average, respectively. While these impacts are remarkably large, they are biologically plausible given the correlations between air pollution (or maternal stress) and birth outcomes found in previous studies.[28] These results are also consistent with those by economists in the context of Pennsylvania for low birth weight: Currie, Greenstone, and Meckel (2017) found an increase of 25 percent within 1 km compared with 3–15 km, Hill (2018) found an increase of 25 percent compared with permitted wells within 2.5 km, and Hill and Ma (2022) found an increase of 11 percent for each additional well drilled within 1 km a public drinking-water source. Given the wealth of studies that identify a causal link between birth weights and long-run outcomes, these impacts are likely to persist throughout these children's lives (Almond and Currie 2011; Almond, Currie, and Duque 2018).

Importantly, the results for maternal risk factors are suggestive of increased stress and exposure to ambient air pollution and are consistent with Willis et al. (2022) in the context of Texas. These results begin to clarify the likely mechanisms that explain the infant health results. The results for different levels of low birth weight and premature birth suggest that oil and gas extraction are not merely increasing those that fall below the threshold, but that there is also an increase in very low birth weight and very premature birth. This suggests that the communities exposed may experience increased infant mortality and certainly higher health-care costs associated with these more vulnerable infants.[29]

It is clear from these results that policies intended to mitigate the risks of OGD can have significant health benefits. I find detectable effects of OGD up to 1.5 km from a well. Current required setback distances (distance between well and nearby residences, hospitals, and schools) range from 300 ft to 800 ft across the 34 states where OGD is taking place.[30] With detectable infant health effects up to 1 mile away, these setback distances may be deemed insufficient to protect human health (see Ericson, Kaffine, and Maniloff [2020] for a discussion of costs associated with increasing setback distances in Colorado). These findings also add impetus for regulators to increase regulations that reduce air pollution emissions from drilling operations and for industry actors to increase voluntary action to reduce air pollution emissions.

A. CONTEXT IN EXISTING LITERATURE

Concurrently to this study, a few other studies have been published on the reproductive health effects of natural gas development. The results reported in this paper are fairly similar to those found in Pennsylvania (Currie, Greenstone, and Meckel 2017; Hill 2018; Hill and Ma 2022) and Texas (Willis et al. 2021, 2022). Studies in Pennsylvania have focused

28 For example, Zahran et al. (2012) found that exposure to benzene (one of the likely contaminants associated with oil and gas development) reduced birth weight by 16.5 grams and increased the odds of a very low birth weight event by a multiplicative factor, and Slama et al. (2009) found that exposure to benzene reduced birth weight by 77 grams.

29 Very low birth weight and very premature birth are major risk factors for infant mortality.

30 COGCC increased the setback from 300 to 500 ft for occupied buildings in 2013. Debate continues in Colorado about extending setbacks for schools or hospitals to 1,000 ft, and some have lobbied for 1,500–2,500 ft.

on shale gas development, whereas my study expands to other types of drilling in Colorado. Given the similarity in results, the findings in this paper suggest that any and all forms of oil and gas extraction, not just shale gas development, are likely to result in adverse birth outcomes.

In contrast to the findings in Pennsylvania and Texas, and the findings in this paper, McKenzie et al. (2014) found that maternal residences near the largest number of natural gas wells within 10 miles in rural Colorado had reduced prevalence of low birth weight and premature birth (i.e., a protective effect), but found an increased risk for congenital heart defects using cross-sectional variation from 1996 to 2009. In this paper, I show that proximity to drilling is correlated with higher socioeconomic status and that mothers living within 1 km boast heavier babies and less prematurity prior to drilling, which likely explains the association in McKenzie et al. (2014). I do not find an increased risk of congenital birth defects and find an increase in low birth weight and premature birth. I believe the discrepancy in findings is easily explained by the difference in empirical approaches.

VIII. Conclusion

Despite substantial policy attention and significant increases in drilling activity across the country, there is little guidance on how environmental and health policy should be designed to protect the health of those citizens who live in close proximity to well development, waste treatment facilities, or shale gas supply routes. This paper highlights the importance of protecting reproductive health for women living near OGD activities, and the importance of studying the health effects for individuals living near conventional development activities, not just shale gas.

As a first step, I assembled with the help of CDPHE a unique data set with the latitude and longitude of new mothers and the locations of oil and gas wells in Colorado and calculated spatiotemporal relationships for all births in Colorado. I examine the impacts of living in close proximity to development on low birth weight, birth weight, and premature birth using a difference-in-differences estimation strategy. Using detailed data on the locations of oil and gas wells in Colorado and the dates they are drilled, I estimate that, on average, mothers living within 1 km of a well prior to birth have reduced birth weight babies and increased prevalence of low birth weight. The result for low birth weight is large, indicating an increase relative to the mean of 31 percent. I also find increased prevalence of premature birth and reductions in gestation lengths in these communities. I provide new evidence on mechanisms, showing that newer drilled wells and coalbed methane wells have the largest impacts on infant health. I also address threats to identification by understanding how OGD in Colorado leads to changes in community composition, migration, and fertility.

While the economic benefits of OGD are easily quantifiable, there may be some hidden external costs. Given that low birth weight and premature birth are strong predictors of education, labor force participation, reduced earnings, and future health, the long-term costs could be very high for these communities. Further research to inform policy makers about the precise mechanisms of exposure is sorely needed, and more research on the health impacts to children and adults, as well as longer-term impacts, is certainly warranted.

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